

Suraj Chougule

Pune, India — +91 8446364651 — surajchougule378@gmail.com

[GitHub](#) — [LinkedIn](#) — [Portfolio](#)

SUMMARY

Software Engineering undergraduate building and deploying **AI-native, full-stack systems** with hands-on experience across Python, JavaScript, React, Node.js, REST APIs, databases, and cloud services. Built production applications integrating **Gemini and Cloudflare AI**, with a focus on end-to-end ownership, measurable performance, reliability, debugging, secure APIs, and iterative engineering. Experienced in turning product requirements into shipped systems, load testing real workloads, analyzing **latency, throughput, failure modes**, and hardening applications for real-world usage. Strong interest in developer tooling, AI-powered engineering workflows, scalable infrastructure, and practical automation.

EDUCATION

International Institute of Information Technology (I²IT), Pune
B.E. Information Technology

Aug 2023 – May 2027
CGPA: 8.84/10

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, C++, C, SQL, HTML5, CSS3

AI/GenAI: Gemini API, Gemini Vision, Cloudflare AI, Generative AI, AI API Integration, Prompt-Based Workflows, AI Middleware

Frontend: React.js, Vite, Tailwind CSS, React Router, Recharts

Backend: Node.js, Express.js, Spring Boot, REST APIs, API Design, Asynchronous Programming, Authentication, Authorization

Databases: PostgreSQL, MongoDB, Redis, SQL, NoSQL, Database Indexing, Query Optimization, ACID Transactions, Audit Logging

Cloud & Infrastructure: AWS, Docker, Cloudflare R2, Vercel, Render, Redis/Upstash, Linux

Testing & Quality: Jest, Supertest, JUnit, Mocha, Postman, Unit Testing, Integration Testing, API Testing, Load Testing, Automated Testing

Engineering: System Design, Distributed Systems, Scalability, Reliability, Fault Tolerance, Performance Optimization, Debugging, Error Handling, Secure Coding, CI/CD, Git, Agile, SDLC

PROJECTS

NoteShare — Production Student Platform

[Live](#) — [GitHub](#)

- Built and deployed an end-to-end platform using **React, Node.js, Express, MongoDB, Redis, and Cloudflare R2**, serving **232+ organic users** with authentication, search, note workflows, analytics, and administrative features.
- Designed production **REST APIs** with JWT authentication, Google OAuth 2.0, RBAC, secure cookies, input validation, rate limiting, and structured error handling.
- Implemented a three-level caching architecture using **NodeCache, Redis, and browser/CDN caching**, reducing repeated database lookups by approximately **90%**.
- Diagnosed and optimized API bottlenecks, reducing **P95 latency from 315ms to 4ms** while improving system throughput and responsiveness.
- Load-tested real workloads using Grafana k6, achieving **1,500+ RPS with 100 concurrent users and 0% request failures**, validating performance and reliability under sustained load.
- Applied database indexing, automated testing, CI/CD deployment, and secure API design to improve **scalability, maintainability, reliability, and production readiness**.

Aura AI — Multilingual Text-to-Image Generator

[Live](#) — [GitHub](#)

- Built and deployed an **AI-native workflow** using React, Node.js, Express, MongoDB, Gemini API, and Cloudflare AI for multilingual text-to-image generation across **11 Indian languages**.
- Designed an end-to-end pipeline of **input processing → AI translation → image generation → persistent user history**, integrating multiple external AI services through backend APIs.
- Implemented asynchronous API workflows, JWT-protected routes, rate limiting, validation, and **fault-tolerant fallback mechanisms** to handle external service failures and request constraints.
- Debugged and iterated across frontend, backend, AI-service integrations, and database workflows to deliver a usable deployed application rather than a standalone prototype.

EquiSplit — Financial Workflow & API Platform

[Live](#) — [GitHub](#)

- Built a production-oriented expense platform using **Node.js, Express, PostgreSQL, Redis, React, and Gemini Vision**, supporting financial workflows, group management, expense processing, and receipt OCR.
- Implemented **ACID-safe transactions, integer-based monetary storage, idempotency, concurrency controls, audit logging, and Redis rate limiting** for reliable financial operations.
- Developed equal, exact, and percentage-based expense workflows with optimized debt settlement using a **Priority Queue with $O(N \log N)$ complexity**.
- Created **100+ automated tests** covering authentication, groups, expenses, settlement algorithms, concurrency, database constraints, idempotency, and rate limiting.

LEADERSHIP

Vice Chairperson – Information Technology Student Association (ITSA), I²IT Pune

Jul 2025 – Aug 2026

- Led and mentored a **50+ member** student team, coordinating responsibilities, requirements, deliverables, and execution across technical and organizational initiatives.
- Coordinated initiatives involving **200+ participants**, strengthening communication, collaboration, problem solving, and ownership across teams.

ACHIEVEMENTS

- **1st Place – Dhruva Nexus Hackathon:** Built an internship portal featuring automatic text extraction and an end-to-end application workflow.